Fiscal Year:	FY 2005	**Task Last Updated:**	FY 11/04/2005
PI Name:	Kennedy, Ann R. D.Sc.		
Project Title:	Countermeasures for Space Radiation Biological Effects		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Radiation Effects Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	akennedy@mail.med.upenn.edu	**Fax:**	FY 215-898-0090
PI Organization Type:	UNIVERSITY	**Phone:**	215-898-0079
Organization Name:	University of Pennsylvania		
PI Address 1:	Department of Radiation Oncology		
PI Address 2:	Oncology Research Division		
PI Web Page:			
City:	Philadelphia	**State:**	PA
Zip Code:	19104-6072	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NSBRI
Start Date:	10/01/2001	**End Date:**	05/31/2005
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			
	It is our hypothesis that control of radiation induced oxidative stress will reduce the risk of cancer development. In this grant work, there are two major specific aims: to determine whether certain dietary supplements can reduce space radiation induced oxidative stress in cultured human cells, and to determine whether the dietary supplements shown to reduce oxidative stress in cultured cells have effects on space radiation induced oxidative stress in animal model systems. The supplemental agents evaluated alone and in combinations include: vitamins C, E, folic acid, glutathione, N-acetyl cysteine (NAC), selenomethionine (SeM), alpha-lipoic acid, niacin, thiamin, Co-enzyme Q10 and the soybean-derived Bowman-Birk inhibitor. The levels of oxidative stress in cultured cells are determined by a dichlorofluorescein fluorescence assay, which has been adapted for evaluation of the effects of space radiations as part of this project. In animals, the total antioxidant status (TAS) in serum or plasma samples is used as the indicator of oxidative stress. The agents shown to reduce space radiation induced oxidative stress in vitro (in MCF10 cells, a human breast epithelial cell		

Task Description:	line) include: vitamin C, vitamin E succinate, NAC, SeM, alpha-lipoic acid (in the reduced form) and Co-enzyme Q10. The effects of treatment with this combination of dietary supplement agents on bio-reduction capacity in animals were evaluated in rats irradiated with iron ions, protons or gamma-rays at a dose of 200 cGy. Sham-irradiated animals were used as controls. The results demonstrated that the TAS decreased in the irradiated animals fed with the control diet. The control animal diet (AIN-93G) contained many of the nutritional supplements at levels comparable on a weight basis to the human RDA levels. Supplementation of the diet with the nutritional supplements prevented the drop in the serum or plasma TAS after the radiation exposure. These results suggest that diet supplementation was effective in preventing the reduction in TAS in the animals irradiated with all three types of radiation. These results indicated that both serum and plasma samples are suitable for a determination of TAS in the irradiated animals. Our results indicate that the dietary supplement agents evaluated can reduce the levels of radiation induced oxidative stress in both in vitro and in vivo systems.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	There are many individuals on earth who are exposed to higher than normal background radiation levels. These would include individuals who are occupationally exposed to radiation, such as workers in the nuclear power industry, those employed in universities and hospitals who work with radiation, those treated with radiation as part of medical procedures, airline flight crews and those who frequently travel by air, etc. The countermeasures developed as part of this NSBRI-funded research could greatly benefit these individuals as well as those involved in space travel.
Task Progress:	During this last funded year, there has been great progress in bringing the work of this grant to the state of being published in peer-reviewed journals. The specific publications are listed below. In the past year, data were analyzed on the efficiency of various types of low and high linear energy transfer (LET) radiations to measure oxidative stress in a standardized dichlorofluorescein (DCF) fluorometric assay. In this system, the oxidation of 2'7'-dichlorofluorescin (DCFH) to 2'7'-dichlorofluorescein, a fluorescent DCFH oxidation product, is measured. In these studies, protons were found to be about equally effective as X-rays in the generation of oxidative stress in cultured cells. However, iron ion beams with energies of 1 GeV/nucleon and 5 GeV/nucleon were less effective than x-rays or gamma rays in inducing DCFH oxidation. The relatively lower slope values for the dose responses of HZE-particle radiation-induced DCFH oxidation indicate that the sensitivity of the DCF fluorometric assay is probably dependent on the LET of the radiation beam. Previous studies performed as part of this grant work indicated that the total antioxidant status measured in the serum or plasma from animals could serve as an indicator of radiation induced oxidative stress. In the past year, there have been several studies performed to attempt to adapt the DCF fluorometric assay for the measurement of radiation induced oxidative stress in solid tissue(s). Much of this work was performed in thyroid tissue, so that results obtained could be directly compared with results from in vitro studies performed in HTori-3 cells, a line of human thyroid cells, used in studies on malignant transformation in this project. In these studies, the thyroid extract's protein quantity was used for normalization of the signal. These studies indicate that the DCF assay system may be suitable for measurement of radiation induced oxidative stress in vivo in various organ systems. In "Journey to Mars", A Whyfiles Feature Presentation, our work was featured on the cover as "Chapter 3: Blasted by the Ray Gun".
Bibliography Type:	Description: (Last Updated: 08/24/2020)
Articles in Peer-reviewed Journals	Wan XS, Bloch P, Ware JH, Zhou Z, Donahue JJ, Guan J, Stewart J, Kennedy AR. "Detection of oxidative stress induced by low- and high-linear energy transfer radiation in cultured human epithelial cells." Radiat Res. 2005 Apr;163(4):364-8. PMID: 15799690 , Apr-2005
Articles in Peer-reviewed Journals	Wan XS, Zhou Z, Ware JH, Kennedy AR. "Standardization of a fluorometric assay for measuring oxidative stress in irradiated cells." Radiat Res. 2005 Feb;163(2):232-40. PMID: 15658900 , Feb-2005
Articles in Peer-reviewed Journals	Wan XS, Ware JH, Zhou Z, Donahue JJ, Guan J, Kennedy AR. "Protection against radiation-induced oxidative stress in cultured human epithelial cells by treatment with antioxidant agents." Int J Radiat Oncol Biol Phys. 2006 Apr 1;64(5):1475-81. PMID: 16472936 , Apr-2006
Awards	Kennedy AR. "NASA Spotlighted Investigator , 2004." Jan-2004